Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

National University of Singapore **2022–present**
Postdoc under Kostya Novoselov

- **Transformer for generating symmetric crystals** :
  - Created a generative model for material design based on symmetry inductive bias.
  - Implemented the model in PyTorch.
  - Achieved the best generation quality and diversity among space-group-conditioned models.
  - Led a team of 6.
- **ML for predicting properties of defects in 2D crystals** :
  - Created a sparse representation of crystals with defects, achieving 4x energy prediction error reduction.
  - Developed code for reproducible parallel running of machine learning experiments.
  - Led a team of 3.

Constructor Group **February 2021–present**
Machine Learning Consultant

- **AI/ML Consulting** :
  - Consulted on machine learning algorithms, deep learning applications, and architectural designs for scientific and industry-led projects.

CERN [Yandex → HSE University] **2014–2022**
Researcher
2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.

- **Generative models for fast simulation** :
  - Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
  - Achieved approximately 10^5 speed-up compared to the ab-initio simulator.
- **Machine learning on noisy data** :
  - Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.
- **Muon identification at the LHCb experiment at CERN** :
  - Solved a classification problem over tabular data with noisy labels under timing constraints.
  - Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
  - Integrated the model into LHCb production, mostly in C++.
  - Packaged the work as a data science competition problem for IDAO-2019.

Education

Higher School of Economics (HSE University) **2016–2020**
PhD in Computer Science, supervisor Andrey Ustyuzhanin; Machine Learning for particle identification in the LHCb detector.

Sapienza — Universita di Roma **2016–2020**
PhD in Physics (double degree with HSE), supervisor Barbara Sciascia.

Product Management course at Yandex **Spring 2018**
Focused coursework in product strategy and execution.

Yandex School of Data Analysis **2013–2015**
Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology

MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.

BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

Mentorship

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 20).
- Co-PI of a $3.4m AI Singapore grant on multiscale machine learning.

Teaching & Outreach

- Pitch research to the general public through major newspaper coverage (1, 2), blog posts (1, 2), a science slam, interviews (1, 2), and a public talk.
- Taught at the Machine Learning in High Energy Physics Summer School in 2015, 2016, 2017, 2018, 2019, 2020, 2021, and 2023.
- Lecturer in the Coursera Advanced Machine Learning specialisation, which is award-winning.
- Delivered more than 20 conference talks.

Service

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

Skills

ML & AI (Structured, non-text, domain-specific data) ·
Python (Research coding) ·
PyTorch (Deep learning) ·
C++ (Production) ·
Linux (Administration) ·
Research Writing & Speaking ·
Physics (Deep domain understanding) ·
Leadership & Mentorship

Selected Publications

1. **Kazeev, Nikita**, and Ian Babich. Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments. OpenReview, June 2026.
2. Artem Dembitskiy, Shuya Yamazaki, Artem Maevskiy, **Nikita Kazeev**, Roman A. Eremin, Semen Budennyy, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials. ICML 2026 AI for Science Workshop, May 2026.
3. Nian Liu, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Yuwei Zeng, Ryoji Kubo, Pengru Huang, Thomas Laurent, Yann LeCun, Kostya S. Novoselov, and Xavier Bresson. Crys-JEPA: Accelerating Crystal Discovery via Embedding Screening and Generative Refinement. arXiv preprint arXiv:2605.14759, May 2026.
4. Nian Liu, Yuwei Zeng, Ryoji Kubo, **Nikita Kazeev**, Stephen Gregory Dale, Artem Maevskiy, Pengru Huang, Thomas Laurent, Kostya S. Novoselov, and Xavier Bresson. Composable Crystals: Controllable Materials Discovery via Concept Learning. arXiv preprint arXiv:2605.14769, May 2026.
5. Jose Recatala-Gomez, **Nikita Kazeev**, et al. Generative design of inorganic materials. arXiv preprint arXiv:2604.14082, April 2026.
6. Alexandre Duval, Siddharth Betala, Samuel P. Gleason, Andy Xu, Georgia Channing, Daniel Levy, Ali Ramlaoui, Clémentine Fourrier, Chaitanya K. Joshi, **Nikita Kazeev**, Sékou-Oumar Kaba, Félix Therrien, Alex Hernández-García, Rocío Mercado, N M Anoop Krishnan. LeMat-GenBench: Bridging the gap between crystal generation and materials discovery. AI for Accelerated Materials Design - NeurIPS 2025 / arXiv preprint arXiv:2512.04562, October 2025.
7. Andrey Okhotin, Maksim Nakhodnov, **Nikita Kazeev**, Mikhail Lazarev, Andrey E. Ustyuzhanin, and Dmitry Vetrov. MiAD: Mirage Atom Diffusion for De Novo Crystal Generation. arXiv preprint arXiv:2511.14426, November 2025.
8. Stephen G. Dale, **Nikita Kazeev**, et al. AI4X Roadmap: Artificial Intelligence for the advancement of scientific pursuit and its future directions. arXiv preprint arXiv:2511.20976, November 2025.

9. M. Y. Lukianov, A. Maevskiy, **N. Kazeev**, D. Mylnikov, D. A. Svintsov, K. S. Novoselov, A. Ustyuzhanin, and D. A. Bandurin. Inverse design of broadband antennas for terahertz devices based on two-dimensional materials. *Physical Review Applied* 24, 054079, November 2025.
10. **Kazeev, Nikita**, et al. WyckoffTransformer: Generation of Symmetric Crystals. ICML 2025, May 2025.
11. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. Sparse representation for machine learning the properties of defects in 2D materials. npj Comput Mater 9, 113 (2023).
12. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). Cherenkov detectors fast simulation using neural networks. *Nuclear Instruments and Methods in Physics Research Section A. I'm the corresponding author.*